

THE DIMENSIONS OF A RING AND ITS FORMAL POWER SERIES RING

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ABSTRACT. Understanding the relation between $\dim R$ and $\dim R[[x]]$ is a classical problem in commutative algebra. For a Noetherian ring R , one has $\dim R[[x]] = \dim R + 1$, but the general case is considerably more delicate. Arnold [1] proved that finite power-series dimension requires the strong finite type (SFT) condition, whereas Coykendall [6] constructed a one-dimensional SFT domain whose power series ring has infinite dimension. The question of Coykendall and Gilmer whether $\dim R[[x]] < \infty$ forces $\dim R[[x]] \leq 2 \dim R + 1$ was answered negatively by Kang and Park [6, 7]. In this paper, we prove that, as R ranges over the nonzero commutative rings with identity, the finite pairs $(\dim R, \dim R[[x]])$ are exactly $(0, 1)$ and the pairs (n, m) with $1 \leq n < m$.

1. BACKGROUND, RELATED RESULTS, AND THE MAIN THEOREM

Throughout, rings are nonzero, commutative, and have an identity, and dimension means Krull dimension. The notation $R[[x]]$ denotes the ring of formal power series in one indeterminate over R . For an ideal $I \subseteq R$, we write

$$I[[x]] = \left\{ \sum_{j \geq 0} a_j x^j : a_j \in I \right\}.$$

This is the kernel of $R[[x]] \rightarrow (R/I)[[x]]$; it need not equal the extended ideal $IR[[x]]$. We write $\mathbb{N} = \{1, 2, \dots\}$ and $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.

The dimension theory of formal power series differs substantially from that of polynomial rings. For a finite-dimensional ring R , Seidenberg's bounds give

$$\dim R + 1 \leq \dim R[x] \leq 2 \dim R + 1 \quad [11].$$

The lower bound also holds for $R[[x]]$, and equality holds when R is Noetherian. In general, however, $R[[x]]$ can have infinite dimension, and even its finite dimension need not satisfy the polynomial upper bound.

An ideal I is of *strong finite type* (SFT) if there exist a finitely generated ideal $J \subseteq I$ and an integer $e \geq 1$ such that $a^e \in J$ for every $a \in I$. A ring is SFT if each of its ideals is SFT. Arnold proved that

$$\dim R[[x]] < \infty \implies R \text{ is SFT} \quad [1]; \quad (1.1)$$

see also Roitman [10]. Condo, Coykendall, and Dobbs [5, Corollary 2.2] proved that, for a zero-dimensional SFT ring R ,

$$\dim R[[x_1, \dots, x_s]] = s \quad (s \geq 1). \quad (1.2)$$

These results account for the exceptional finite pair $(0, 1)$.

Several important classes have more restrictive dimension formulas. For a finite-dimensional Prüfer domain D , Arnold's theorem gives $\dim D[[x]] = \dim D + 1$ precisely when D is SFT, and

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infinite dimension otherwise [2, Theorem 3.8]. More generally, if D is an SFT Prüfer domain of positive finite dimension d , then

$$\dim D[[x_1, \dots, x_s]] = sd + 1 \quad [4, \text{Theorem 3.6}].$$

The SFT hypothesis alone does not ensure finite power-series dimension: Coykendall [6] constructed a one-dimensional SFT domain whose power series ring has infinite dimension.

Arnold [3] had already shown that a finite power-series dimension need not equal the coefficient dimension plus one. Kang and Park [7] constructed finite-dimensional counterexamples to the bound $\dim R[[x]] \leq 2 \dim R + 1$. Their mixed-extension formula yields, for suitable SFT Prüfer domains D of dimension d and $R = D[z_1, \dots, z_p]$,

$$(\dim R, \dim R[[x]]) = (d + p, d(p + 1) + 1).$$

For example, $d = 4$ and $p = 3$ give $(7, 17)$. Thus finite violations of the polynomial bound are established prior results. The construction below instead keeps the coefficient dimension equal to one and prescribes every finite power-series dimension at least two.

Our method also relates to the study of subrings of power series rings defined by bounds on the degrees of their coefficients; see Kang and Toan [8, 9]. Here the coefficients are rational functions, and the filtration controls both their denominators and their numerator degrees. We prove the required division estimate and Noetherianity directly.

Theorem 1.1. *Let $n, m \in \mathbb{N}_0$. There exists a nonzero commutative ring R with identity such that*

$$\dim R = n \quad \text{and} \quad \dim R[[x]] = m$$

if and only if

$$(n, m) = (0, 1) \quad \text{or} \quad 1 \leq n < m.$$

The essential realization statement is stronger in dimension one.

Theorem 1.2. *For every integer $m \geq 2$, there exists a one-dimensional local domain T such that $\dim T[[x]] = m$.*

For $m = 2$ one may take $T = k[[c]]$, where k is a field. For $m = r + 2 \geq 3$, Sections 2–4 construct a local domain T_r and prove

$$\dim T_r = 1, \quad \dim T_r[[x]] = r + 2. \quad (1.3)$$

Section 5 deduces Theorem 1.1. Its higher-dimensional realizations use finite direct products; the theorem is stated for rings without a domain hypothesis.

2. THE COEFFICIENT FILTRATION AND THE LOCAL DOMAIN

Fix $r \geq 1$ and let $k = \overline{\mathbb{Q}}$. Let $u_1, \dots, u_r$ be algebraically independent indeterminates over k . Write $U = (u_1, \dots, u_r)$ and $k[U] = k[u_1, \dots, u_r]$, and let $K = k(U)$ be the fraction field of $k[U]$. Since $k[U]$ is countable, we may enumerate representatives of its associate classes of irreducible polynomials as $q_1, q_2, \dots$. All polynomial degrees below are total degrees in U ; we set $\deg 0 = -\infty$. For $N \in \mathbb{N}_0$, define

$$D_N = \prod_{j=1}^N q_j^{\lfloor N/j \rfloor}, \quad W_N = \left\{ \frac{p}{D_N} : p \in k[U], \deg p \leq \deg D_N + N \right\}. \quad (2.1)$$

Thus $D_0 = 1$ and $W_0 = k$.

Lemma 2.1. *The spaces W_N are finite-dimensional over k and satisfy*

$$W_N \subseteq W_{N+1}, \quad W_N W_M \subseteq W_{N+M}, \quad \bigcup_{N \geq 0} W_N = K. \quad (2.2)$$

Consequently the function $w(a) = \min\{N : a \in W_N\}$ on K has $w(0) = 0$ and satisfies

$$w(a+b) \leq \max\{w(a), w(b)\}, \quad w(ab) \leq w(a) + w(b). \quad (2.3)$$

Proof. Finite dimensionality follows from the numerator degree bound. The inclusions follow from

$$D_N \mid D_{N+1}, \quad D_N D_M \mid D_{N+M},$$

the second divisibility being a consequence of $\lfloor N/j \rfloor + \lfloor M/j \rfloor \leq \lfloor (N+M)/j \rfloor$. For example, after writing a product over D_{N+M} , its numerator has degree at most

$$(\deg D_N + N) + (\deg D_M + M) + \deg \frac{D_{N+M}}{D_N D_M} = \deg D_{N+M} + N + M.$$

For $p/q \in K$, choose N such that $q \mid D_N$ and $N \geq \deg p - \deg q$. Then $p/q \in W_N$, proving exhaustion. The inequalities for w follow from the vector-space and multiplicative properties of the filtration. $\square$

Set $V_j = W_{2^j-1}$ and define a subring of $K[[c]]$ by

$$T = T_r = \left\{ \sum_{j \geq 0} a_j c^j : a_j \in V_j \right\}. \quad (2.4)$$

Indeed, $V_i V_j \subseteq V_{i+j}$ because $(2^i - 1) + (2^j - 1) \leq 2^{i+j} - 1$. In particular T is a domain, $c \in T$, and every constant coefficient belongs to k .

We will use the elementary estimate

$$\sum_{i=1}^s 2^{j_i} \leq 2^n \quad \text{if } j_i \geq 1 \text{ and } \sum_{i=1}^s j_i = n. \quad (2.5)$$

It follows by repeatedly applying $2^a + 2^b \leq 2^{a+b}$ for $a, b \geq 1$. Hence, in a ring with a weight satisfying (2.3) and $w(1) = w(-1) = 0$, the condition $w(b_j) \leq C2^j$ for $j \geq 1$ implies that the coefficients d_n in

$$\left(1 + \sum_{j \geq 1} b_j c^j \right)^{-1} = \sum_{n \geq 0} d_n c^n \quad (2.6)$$

satisfy $w(d_n) \leq C2^n$. Indeed, $d_0 = 1$, and for $n \geq 1$,

$$d_n = \sum_{s=1}^n (-1)^s \sum_{\substack{j_1 + \dots + j_s = n \\ j_i \geq 1}} b_{j_1} \cdots b_{j_s}.$$

Proposition 2.2. *The ring T is local, with maximal ideal*

$$\mathfrak{m} = \left\{ \sum_{j \geq 0} a_j c^j \in T : a_0 = 0 \right\}.$$

Moreover,

$$\mathfrak{m}^3 \subseteq cT, \quad \sqrt{cT} = \mathfrak{m}, \quad T[1/c] \text{ is a field}, \quad \text{Spec } T = \{(0), \mathfrak{m}\}. \quad (2.7)$$

In particular, $\dim T = 1$.

Proof. If $a \in T$ has nonzero constant coefficient, scale that coefficient to one. The expansion in (2.6) and $V_{j_1} \cdots V_{j_s} \subseteq V_{j_1 + \dots + j_s}$ show that its formal inverse belongs to T . Thus T is local and $T/\mathfrak{m} = k$.

For positive i, j, ℓ with $i + j + \ell = n$, one has

$$(2^i - 1) + (2^j - 1) + (2^\ell - 1) \leq 2^{n-2} + 1 \leq 2^{n-1} - 1 \quad (n \geq 3). \quad (2.8)$$

For the first inequality, shifting one unit from a smaller index greater than one to a largest index does not decrease the sum of the powers of two; the maximum is attained at $(n-2, 1, 1)$. Every coefficient of c^n in a product of three elements of $\mathfrak{m}$ therefore belongs to V_{n-1} . Division by c gives $\mathfrak{m}^3 \subseteq cT$, and $\sqrt{cT} = \mathfrak{m}$ follows.

If $w(b_j) \leq C2^j$ for all $j \geq 0$, then

$$c^d \sum_{j \geq 0} b_j c^j \in T \quad \text{whenever } 2^d \geq C + 1, \quad (2.9)$$

since $C2^j \leq 2^{j+d} - 1$. For $0 \neq a \in T$, write

$$a = c^\nu a_\nu \left(1 + \sum_{j \geq 1} b_j c^j \right), \quad b_j = a_{\nu+j} a_\nu^{-1}, \quad a_\nu \neq 0.$$

The estimate $w(b_j) \leq 2^{\nu+j} - 1 + w(a_\nu^{-1})$ gives an exponential bound. Formula (2.6) gives the same type of bound for the reciprocal of the parenthesized series; multiplication by the fixed scalar a_ν^{-1} preserves it. Equation (2.9), with $c^{-\nu}$ absorbed into the shift, yields $c^d/a \in T$ for some d . Thus every nonzero element of T is invertible in $T[1/c]$.

Primes avoiding c correspond to primes of this field, so the only such prime is (0) . The radical identity shows that $\mathfrak{m}$ is the only prime containing c . Since $0 \neq c \in \mathfrak{m}$, the spectrum and dimension assertions follow. $\square$

3. BOUNDED DIVISION AND AN AUXILIARY NOETHERIAN RING

Put

$$F = k[[x]], \quad L = k((x)), \quad A_0 = F[U], \quad S = k[U] \setminus \{0\}, \quad B = S^{-1}A_0. \quad (3.1)$$

The natural embeddings identify B with $K \otimes_k F$ inside $K[[x]]$. The ring B is a Noetherian domain, and

$$\dim B \leq r. \quad (3.2)$$

Indeed, if $\mathfrak{q} \in \text{Spec } B$ avoids x , its lower prime chains lie in $B[1/x]$, a localization of $L[U]$, and hence $\text{ht } \mathfrak{q} \leq r$. If $x \in \mathfrak{q}$, then $B/xB = K$ gives $\mathfrak{q} = xB$. The principal ideal theorem gives $\text{ht}(xB) = 1 \leq r$.

Extend the filtration to B by

$$E_N = W_N \otimes_k F = \left\{ \frac{p}{D_N} : p \in F[U], \deg p \leq \deg D_N + N \right\}. \quad (3.3)$$

Under the embedding into $K[[x]]$, one also has

$$W_N[[x]] = E_N. \quad (3.4)$$

Here the left side means series whose coefficients lie in W_N . To verify the identity, expand each coefficient in the finite k -basis $\{U^\alpha/D_N : |\alpha| \leq \deg D_N + N\}$ and collect the coefficients of each basis element into a series in F . The E_N exhaust B , are increasing and multiplicative, and define $w(b) = \min\{N : b \in E_N\}$ on B . The inequalities (2.3) still hold, and every element of F has weight zero.

Lemma 3.1 (Bounded division). *Fix $b_1, \dots, b_s \in B$ and let $J = (b_1, \dots, b_s)B$. There is an integer $C \geq 0$ such that, for every $N \geq 0$ and $h \in J \cap E_N$, one can write*

$$h = \sum_{i=1}^s b_i z_i, \quad z_i \in E_{N+C}. \quad (3.5)$$

Proof. The case $J = 0$ is immediate. Set $J_0 = J \cap A_0$ and filter $A_0 = F[U]$ by total degree in U . Its associated graded ring is $F[U]$, which is Noetherian. Thus the homogeneous ideal generated by the highest-degree parts of elements of J_0 is generated by the highest-degree parts of finitely many nonzero elements $\ell_1, \dots, \ell_t \in J_0$. Write $d_j = \deg \ell_j$. Every $0 \neq a \in J_0$ admits a representation

$$a = \sum_{j=1}^t \ell_j p_j, \quad p_j \in F[U], \quad \deg p_j \leq \deg a - d_j. \quad (3.6)$$

Indeed, express the highest-degree part of a using homogeneous multipliers of degree $\deg a - d_j$, subtract the corresponding combination of the ℓ_j , and repeat at the smaller degree. This process terminates. A negative degree bound means $p_j = 0$; no division by a nonunit of F is involved.

Since $\ell_j \in J$, there are fixed $s_0 \in S$ and $A_{ij} \in F[U]$ with

$$\ell_j = \sum_{i=1}^s b_i \frac{A_{ij}}{s_0} \quad (1 \leq j \leq t).$$

For $h \in J \cap E_N$, write $h = a/D_N$, where $a \in A_0$ and $\deg a \leq \deg D_N + N$. Since D_N is a unit of B , we have $a \in J_0$. Using (3.6), set

$$z_i = \frac{\sum_j A_{ij} p_j}{s_0 D_N}; \quad h = \sum_i b_i z_i.$$

There is a fixed $C_0 \geq 0$ such that each numerator here has degree at most $\deg D_N + N + C_0$.

Choose C so large that

$$s_0 D_N \mid D_{N+C} \quad (N \geq 0), \quad C \geq C_0 - \deg s_0. \quad (3.7)$$

To obtain the divisibility, for each factor $q_j^{e_j}$ of s_0 it suffices to require $C \geq j e_j$, since

$$\left\lfloor \frac{N+C}{j} \right\rfloor - \left\lfloor \frac{N}{j} \right\rfloor \geq e_j.$$

Writing z_i over D_{N+C} now bounds its numerator degree by

$$\deg D_N + N + C_0 + \deg \frac{D_{N+C}}{s_0 D_N} = \deg D_{N+C} + N + C_0 - \deg s_0 \leq \deg D_{N+C} + N + C.$$

Thus $z_i \in E_{N+C}$, as required. $\square$

Inside $B[[c]]$, define

$$H = \left\{ \sum_{n \geq 0} b_n c^n : \text{there is } M \geq 0 \text{ with } w(b_n) \leq M 2^n \text{ for all } n \right\}. \quad (3.8)$$

The bound may depend on the series. This is a subring containing B : for multiplication, the weight of the coefficient of c^n is at most

$$\max_{0 \leq i \leq n} (M 2^i + M' 2^{n-i}) \leq (M + M') 2^n.$$

Shifting the coefficients gives

$$H \cap c^e B[[c]] = c^e H \quad (e \geq 0), \quad H/cH \cong B, \quad \text{gr}_c H \cong B[z], \quad (3.9)$$

where z is the initial form of c . Furthermore,

$$c \in \text{Jac}(H). \quad (3.10)$$

Indeed, for every $h \in H$, the positive-degree coefficients of ch satisfy an exponential bound. Formula (2.6) shows that $(1 - ch)^{-1} \in H$.

Proposition 3.2. *The ring H is Noetherian, and $\dim H[1/c] \leq r$.*

Proof. Let $0 \neq I \subseteq H$ be an ideal. Since $B[z]$ is Noetherian, the ideal of initial forms of I is generated by the initial forms of finitely many nonzero elements $f_1, \dots, f_s \in I$. Write

$$e_i = \text{ord}_c(f_i), \quad f_i = \sum_{j \geq 0} f_{i,e_i+j} c^{e_i+j}, \quad b_i = f_{i,e_i}.$$

For each $n \geq 0$ set $J_n = (b_i : e_i \leq n)B$. There are only finitely many such generator lists, so Lemma 3.1 supplies one constant $C \geq 0$ valid for all the nonzero J_n .

Fix $g = \sum_{n \geq 0} g_n c^n \in I$. Choose $A, F_0 \geq 0$ with

$$w(g_n) \leq A2^n \quad (n \geq 0), \quad w(f_{i,e_i+j}) \leq F_0 2^j \quad (j \geq 1, 1 \leq i \leq s). \quad (3.11)$$

We cancel the coefficients of g successively. After canceling the coefficients below degree n by finitely many multiples of the f_i , the residual still lies in I . Its coefficient h_n at degree n belongs to J_n : if nonzero, its initial form lies in the homogeneous ideal generated by $b_i z^{e_i}$. By bounded division choose $a_{i,n} \in B$ such that

$$h_n = \sum_{e_i \leq n} b_i a_{i,n}, \quad w(a_{i,n}) \leq w(h_n) + C, \quad (3.12)$$

and put $a_{i,n} = 0$ when $e_i > n$. If $h_n = 0$, choose all coefficients to be zero. Subtracting $\sum_{e_i \leq n} a_{i,n} c^{n-e_i} f_i$ cancels degree n as well.

We must verify that the resulting infinite multipliers belong to H . Put $T_n = \max_i w(a_{i,n})$. A cancellation at degree $m < n$ contributes to degree n a product $a_{i,m} f_{i,e_i+n-m}$. Thus (2.3), (3.11), and (3.12) give

$$T_n \leq \max \left\{ A2^n, \max_{0 \leq m < n} (T_m + F_0 2^{n-m}) \right\} + C, \quad (3.13)$$

where the inner maximum is omitted for $n = 0$. Choose $M \geq \max\{A + C, 2F_0 + C\}$. We claim $T_n \leq M2^n$. The case $n = 0$ is immediate. If the claim holds below n , then, for $d = n - m \geq 1$,

$$\begin{aligned} M(2^n - 2^m) &= M2^m(2^d - 1) \\ &\geq (2F_0 + C)(2^d - 1) \geq F_0 2^d + C. \end{aligned}$$

It follows that $T_m + F_0 2^{n-m} + C \leq M2^n$; also $A2^n + C \leq M2^n$. Equation (3.13) proves the induction step. Consequently

$$h_i = \sum_{n \geq e_i} a_{i,n} c^{n-e_i} \in H,$$

because its coefficient of degree j has weight at most $M2^{e_i} 2^j$. The cancellation gives the coefficientwise identity $g = \sum_i f_i h_i$ in $B[[c]]$, and hence in H . Thus $I = (f_1, \dots, f_s)H$. The zero ideal is finitely generated as well, proving that H is Noetherian.

For every maximal ideal $\mathfrak{n}$ of H , (3.10) and the Noetherian local dimension inequality for a principal quotient give

$$\dim H_{\mathfrak{n}} \leq \dim(H_{\mathfrak{n}}/cH_{\mathfrak{n}}) + 1 \leq \dim B + 1 \leq r + 1.$$

Therefore $\dim H \leq r + 1$. Every prime chain avoiding c can be extended strictly by a maximal ideal, since every maximal ideal contains c . Thus every such chain has length at most r , and $\dim H[1/c] \leq r$. $\square$

4. THE DIMENSION OF THE FORMAL POWER SERIES RING

Let $A = T[[x]]$. Interchanging the x - and c -coefficients and using (3.4) identifies A with the subring

$$A = \left\{ \sum_{n \geq 0} b_n c^n \in B[[c]] : b_n \in E_{2^n-1} \text{ for all } n \right\}. \quad (4.1)$$

Indeed, a series in $T[[x]]$ gives at each c -degree n an element of $V_n[[x]] = E_{2^n-1}$. Conversely, expanding each b_n in x gives coefficients in V_n , so each resulting x -coefficient belongs to T . Multiplication agrees because each coefficient of $x^i c^j$ involves only a finite sum.

Proposition 4.1. *One has $\dim A \leq r + 2$.*

Proof. Equation (4.1) gives $A \subseteq H$. If $h = \sum b_n c^n \in H$ has bound $w(b_n) \leq M2^n$, choose d with $2^d \geq M + 1$. Then $M2^n \leq 2^{n+d} - 1$, so $c^d h \in A$. Consequently

$$A[1/c] = H[1/c], \quad \dim A[1/c] \leq r. \quad (4.2)$$

Put $P = \mathfrak{m}[[x]]$. Since $\mathfrak{m}^3 \subseteq cT$,

$$P^3 \subseteq (cT)[[x]] = cA \subseteq P.$$

The equality in the middle follows by dividing coefficients by c . Since $A/P \cong k[[x]]$, we obtain

$$\sqrt{cA} = P, \quad \dim(A/cA) = 1. \quad (4.3)$$

In a finite prime chain of A , the segment avoiding c has length at most r , and the segment containing c has length at most one. If both occur, there is one additional inclusion between them. Thus every such chain has length at most $r + 2$. $\square$

Lemma 4.2. *There exist $f_1, \dots, f_r \in k[[x]]$ algebraically independent over k .*

Proof. Choose $\lambda_1, \dots, \lambda_r \in k$ linearly independent over $\mathbb{Q}$ and set

$$f_i(x) = \exp(\lambda_i x) - 1 = \sum_{n \geq 1} \frac{\lambda_i^n}{n!} x^n.$$

Such λ_i exist: for $r > 1$, a root α of the Eisenstein polynomial $t^r - 2$ gives $1, \alpha, \dots, \alpha^{r-1}$; for $r = 1$ take $\lambda_1 = 1$. A polynomial relation in the f_i , after an invertible polynomial translation, would yield

$$\sum_{\nu} a_{\nu} \exp(\mu_{\nu} x) = 0, \quad \mu_{\nu} = \sum_i \nu_i \lambda_i,$$

with finitely many distinct multi-indices ν and not all $a_{\nu} \in k$ zero. The μ_{ν} are distinct. List them as $\mu_1, \dots, \mu_s$. Formal differentiation at zero in orders $0, \dots, s-1$ gives a homogeneous linear system with determinant

$$\det(\mu_j^{i-1})_{1 \leq i, j \leq s} = \prod_{1 \leq i < j \leq s} (\mu_j - \mu_i) \neq 0,$$

a contradiction. All operations are formal in characteristic zero. $\square$

Proposition 4.3. *The ring A contains a prime chain of length $r + 2$. Thus $\dim T[[x]] = r + 2$.*

Proof. Choose the series in Lemma 4.2 and put

$$\mathfrak{q}_j = (u_1 - f_1(x), \dots, u_j - f_j(x))B \quad (1 \leq j \leq r), \quad \mathfrak{q}_0 = (0).$$

Before localizing, substitution of $f_i(x)$ for u_i , $i \leq j$, defines a surjection

$$F[U] \longrightarrow F[u_{j+1}, \dots, u_r]$$

with kernel $(u_1 - f_1(x), \dots, u_j - f_j(x))F[U]$. Its kernel is disjoint from S : expanding a nonzero element of $k[U]$ in the remaining variables would otherwise give a nonzero polynomial relation over k among $f_1, \dots, f_j$. Localization therefore gives proper prime ideals and a strict chain

$$(0) = \mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \dots \subsetneq \mathfrak{q}_r. \quad (4.4)$$

For strictness, $u_j - f_j(x)$ is nonzero in the domain obtained after the preceding substitution, and remains nonzero after localization. Full substitution extends to $B \rightarrow L$ with kernel $\mathfrak{q}_r$, since every element of S has nonzero image in $F \subseteq L$. This map fixes F , so

$$\mathfrak{q}_r \cap F = (0). \quad (4.5)$$

The coefficientwise ideal $\mathfrak{q}_j[[c]] \subseteq B[[c]]$ is prime, since its quotient is $(B/\mathfrak{q}_j)[[c]]$, a domain. Contract these primes to A :

$$Q_j = \mathfrak{q}_j[[c]] \cap A \quad (0 \leq j \leq r).$$

They all avoid c , and $Q_0 = (0)$. Since $u_j = D_1 u_j / D_1 \in W_1$ and $f_j(x) \in F$, one has

$$c(u_j - f_j(x)) \in Q_j \setminus Q_{j-1} \quad (1 \leq j \leq r),$$

where membership in A follows from (4.1). Thus the contracted chain is strict.

If $a \in Q_r$, its coefficient at c^0 belongs to $E_0 = F$ by (4.1) and to $\mathfrak{q}_r$ by definition. It is zero by (4.5). This says exactly that every x -coefficient of a belongs to $\mathfrak{m}$, so $Q_r \subseteq P = \mathfrak{m}[[x]]$. The inclusion is strict because $c \in P \setminus Q_r$. Finally $P \subsetneq P + (x)$ is a strict prime inclusion, since $A/P \cong k[[x]]$. Hence

$$(0) = Q_0 \subsetneq Q_1 \subsetneq \dots \subsetneq Q_r \subsetneq P \subsetneq P + (x) \quad (4.6)$$

has length $r + 2$. Combining this with Proposition 4.1 proves the dimension equality. $\square$

Proof of Theorem 1.2. For $m = 2$ use the discrete valuation ring $k[[c]]$. For $m \geq 3$ set $r = m - 2$ and use T_r from (2.4). Its required dimensions follow from Propositions 2.2 and 4.3. $\square$

5. THE CLASSIFICATION OF FINITE PAIRS

Proof of Theorem 1.1. Suppose first that $\dim R = n$ and $\dim R[[x]] = m$ are finite. Choose a chain of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n$$

in R . For each i , coefficientwise reduction gives

$$R[[x]]/\mathfrak{p}_i[[x]] \cong (R/\mathfrak{p}_i)[[x]],$$

which is a domain. Thus each coefficientwise ideal $\mathfrak{p}_i[[x]]$ is prime. Their inclusions are strict, as witnessed by constant series from $\mathfrak{p}_i \setminus \mathfrak{p}_{i-1}$. Moreover,

$$R[[x]]/(\mathfrak{p}_n[[x]] + (x)) \cong R/\mathfrak{p}_n$$

is a domain, and $x \notin \mathfrak{p}_n[[x]]$. Consequently

$$\mathfrak{p}_0[[x]] \subsetneq \dots \subsetneq \mathfrak{p}_n[[x]] \subsetneq \mathfrak{p}_n[[x]] + (x)$$

is a prime chain of length $n + 1$, proving

$$m \geq n + 1. \quad (5.1)$$

If $n = 0$, finiteness of m and Equation (1.1) imply that R is an SFT ring. The zero-dimensional result Equation (1.2) then gives $m = 1$. Hence every finite pair has the asserted form.

Conversely, a field realizes $(0, 1)$. Fix integers $1 \leq n < m$. By Theorem 1.2, choose a local integral domain T such that

$$\dim T = 1, \quad \dim T[[x]] = m.$$

Let

$$C = k[z_1, \dots, z_n], \quad R = T \times C.$$

The polynomial ring C is Noetherian of dimension n , so the Noetherian power-series dimension formula yields $\dim C[[x]] = n + 1$. The prime spectrum of a finite direct product is the disjoint union of the spectra of its factors, and coefficientwise projection gives

$$(T \times C)[[x]] \cong T[[x]] \times C[[x]].$$

It follows that

$$\dim R = \max\{1, n\} = n, \quad \dim R[[x]] = \max\{m, n + 1\} = m.$$

This realizes every asserted pair. $\square$

Corollary 5.1. *For every fixed integer $n \geq 1$, the set*

$$\{\dim R[[x]] : \dim R = n, \dim R[[x]] < \infty\}$$

is unbounded. For $n = 1$, it remains unbounded when R is required to be a local integral domain.

Proof. The first assertion follows from Theorem 1.1, and the second follows from Theorem 1.2. $\square$

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